

L001 Universal Login Screen - SRS Specification

SRS Section: 5.1.1 | **Screen ID:** L001 | **Version:** 1.0 | **Status:** Draft **Module:** SharedFoundations (L-series) **Route:** /Login **Source:** [L01 Universal Login.md](#) **Last Updated:** 2026-01-01

1. Screen Overview

1.1 Purpose

The Universal Login Screen (L001) serves as the single authentication entry point for all NewPOPSys web portal users. It provides unified credential-based authentication with role-based routing, multi-factor authentication support, SSO integration for enterprise customers, and password recovery functionality.

1.2 Access Requirements

Attribute	Specification
Authentication	Unauthenticated access required
Pre-conditions	User not currently authenticated
Post-conditions	Authenticated session established, redirect to role-appropriate dashboard
Served Portals	Brand Admin, Store Portal, PSP Operations, Regional Dashboard

1.3 Navigation Path

Entry Point	Path
Direct URL	https://{tenant}.newpopsys.com/login
Session expiry	Automatic redirect from any authenticated route
Logout action	Redirect from any portal
Bookmark/Deep link	Redirect with return URL parameter

2. User Roles & Permissions

2.1 Accessible Roles

All system roles may access this screen prior to authentication. Post-authentication routing is determined by role hierarchy.

Role Enum	Display Name	Post-Login Route
PLATFORM_ADMIN	Platform Admin	/psp/dashboard
PSP_ADMIN	PSP Admin	/psp/dashboard
PSP_OPS	Production Operator / Support Agent	/psp/orders

BRAND_ADMIN	Brand Admin	/admin/dashboard
CAMPAIGN_MANAGER	Campaign Manager	/admin/dashboard
REGIONAL_MANAGER	Regional Manager	/admin/regional
STORE_MANAGER	Store Manager	/store/dashboard
STORE_OPERATOR	Store Operator	/store/campaigns

2.2 Role-Based Routing Priority

REQ-L001-FR-001: The system SHALL route authenticated users to their primary dashboard based on highest-priority role assignment.

Priority Order (highest first):

1. PLATFORM_ADMIN / PSP_ADMIN → /psp/dashboard

2. PSP_OPS → /psp/orders

3. BRAND_ADMIN / CAMPAIGN_MANAGER → /admin/dashboard

4. REGIONAL_MANAGER → /admin/regional

5. STORE_MANAGER → /store/dashboard

6. STORE_OPERATOR → /store/campaigns

2.3 Multi-Role Handling

REQ-L001-FR-002: When a user holds multiple roles across different levels, the system SHALL:

- Use the highest-level role (Tenant > Brand > Store)
- Within the same level, use the highest-permission role
- Display a role selector modal if the user has multiple significant roles

2.4 MFA Requirements by Role

Role	MFA Requirement	Source
PLATFORM_ADMIN	Mandatory (TOTP/WebAuthn)	RBAC-006
PSP_ADMIN	Mandatory (TOTP/WebAuthn)	RBAC-006
BRAND_ADMIN	Mandatory (TOTP/WebAuthn)	RBAC-006
PSP_OPS	Recommended (tenant policy)	RBAC-006
REGIONAL_MANAGER	Recommended (tenant policy)	RBAC-006
CAMPAIGN_MANAGER	Optional (brand policy)	RBAC-006
STORE_MANAGER	Optional (brand policy)	RBAC-006
STORE_OPERATOR	Optional (brand policy)	RBAC-006

3. UI Components

3.1 Layout Structure

REQ-L001-UI-001: The login screen SHALL display a centered card layout with responsive breakpoints.

Breakpoint	Layout	Max Width
Desktop (>1024px)	Centered card with branded background	420px
Tablet (768-1024px)	Centered card, reduced padding	380px
Mobile (<768px)	Full-width, stacked layout	100%

3.2 Primary Components

Component ID	Type	Description	Required
logo	Image	NewPOPSys brand logo (SVG, 120x40px)	Yes
heading	Text	"Welcome Back"	Yes
subheading	Text	"Sign in to your account"	Yes
email-input	Text Field	User email address input	Yes
password-input	Password Field	Password with visibility toggle	Yes
remember-me	Checkbox	"Remember me" session extension	Yes
forgot-password-link	Link	Trigger password reset modal	Yes
login-button	Primary Button	Submit credentials	Yes
sso-divider	Divider	"or" separator	Yes
sso-button	Secondary Button	"Sign in with SSO"	Yes
help-text	Text	"Need help? Contact your administrator"	Yes
error-alert	Alert Banner	Authentication error display	Conditional

3.3 Login Form Wireframe



Password	
+-----+	
*****	[eye]
+-----+	
☒ Remember me	Forgot pass?
+-----+	
Sign In	
+-----+	
----- or -----	
+-----+	
☐ [lock] Sign in with SSO	
+-----+	
+-----+	
Need help? Contact your administrator	
+-----+	

3.4 MFA Modal

REQ-L001-UI-002: When MFA is required, the system SHALL display a modal dialog for code entry.

+-----+	
Two-Factor Authentication	[X]
+-----+	
Enter the 6-digit code from your authenticator app	
+-----+	
_ _ _ _ _ _ _	
+-----+	
☐ Trust this device for 30 days	
+-----+	
Cancel	Verify
+-----+	
Lost access? Use backup code	
+-----+	

3.5 Forgot Password Modal

REQ-L001-UI-003: Password reset SHALL be initiated via a modal dialog.

+-----+	
Reset Password	[X]
+-----+	

```
|
| Enter your email address and we'll
| send you a link to reset your
| password.
|
| Email Address
| +-----+
| | user@company.com |
| +-----+
|
| [Cancel]          [Send Reset Link]
+-----+
```

3.6 SSO Domain Entry Modal

REQ-L001-UI-004: SSO authentication SHALL prompt for company domain.

```
+-----+
| Enterprise Sign In          [X] |
+-----+
|
| Enter your company domain to be
| redirected to your identity
| provider.
|
| Company Domain
| +-----+
| | acmecorp.com |
| +-----+
|
| [Cancel]          [Continue]
+-----+
```

3.7 Role Selector Modal

REQ-L001-UI-005: For multi-role users, the system SHALL display a role selection modal.

```
+-----+
| Select Account              [X] |
+-----+
|
| You have access to multiple
| accounts. Select one to continue:
|
| +-----+
| | [building] Acme Print Services |
| |   PSP Administrator             |
| +-----+
|
| +-----+
| | [tag] Brand Corp               |
| |   Brand Admin                  |
| +-----+
```

	+-----+	
	+-----+	
	[store] Downtown Store #123	
	Store Manager	
	+-----+	
+-----+		

4. Data Requirements

4.1 Entities Involved

Entity	Table	Access Mode	Purpose
User	users	Read	Credential validation
Membership	memberships	Read	Role assignment retrieval
Session	sessions (cache)	Write	Session creation
AuditEvent	audit_events	Write	Login event logging

4.2 User Entity Fields

REQ-L001-DR-001: The following user fields SHALL be accessed during authentication:

Field	Type	Purpose
id	UUID	User identifier
email	VARCHAR(255)	Login credential
password_hash	VARCHAR(255)	bcrypt-hashed password
status	ENUM	Account status (active/suspended/disabled)
mfa_enabled	BOOLEAN	MFA enrollment status
mfa_secret	VARCHAR(255)	TOTP secret (encrypted)
failed_login_count	INTEGER	Lockout counter
locked_until	TIMESTAMPZ	Lockout expiration
is_active	BOOLEAN	Account active flag

4.3 Membership Entity Fields

REQ-L001-DR-002: Role determination SHALL query membership records:

Field	Type	Purpose
user_id	UUID FK	User reference

brand_id	UUID FK	Brand scope (nullable)
role	role_enum	Assigned role
region_scope_id	UUID FK	Regional scope (nullable)

4.4 Session Data Structure

REQ-L001-DR-003: The session object SHALL contain:

```
interface SessionData {
  session_id: string;      // UUID v4
  user_id: string;         // User UUID
  email: string;           // User email
  roles: RoleMembership[]; // Array of role assignments
  primary_role: string;    // Highest-priority role
  tenant_id: string;       // PSP tenant UUID
  brand_id?: string;       // Active brand (if applicable)
  created_at: number;      // Unix timestamp
  expires_at: number;      // Unix timestamp (8 hours default)
  idle_expires_at: number; // Unix timestamp (30 min from last activity)
  mfa_verified: boolean;   // MFA completion status
  remember_me: boolean;    // Extended session flag
  device_fingerprint: string; // Device identifier
}
```

5. Business Rules & Validation

5.1 Input Validation Rules

REQ-L001-BR-001: Email field validation:

Rule	Requirement	Error Message
Required	Field must not be empty	"Email address is required"
Format	Must match RFC 5322 email format	"Please enter a valid email address"
Max Length	255 characters	"Email address is too long"

REQ-L001-BR-002: Password field validation:

Rule	Requirement	Error Message
Required	Field must not be empty	"Password is required"
Min Length	8 characters (display validation only)	"Password must be at least 8 characters"

REQ-L001-BR-003: MFA code validation:

Rule	Requirement	Error Message
------	-------------	---------------

Required	Field must not be empty	"Verification code is required"
Format	Exactly 6 numeric digits	"Please enter a 6-digit code"

REQ-L001-BR-004: SSO domain validation:

Rule	Requirement	Error Message
Required	Field must not be empty	"Company domain is required"
Format	Valid domain format	"Please enter a valid domain"

5.2 Authentication Business Rules

REQ-L001-BR-005: Account lockout policy:

Threshold	Action	Duration
3 failures	CAPTCHA challenge required	Until successful CAPTCHA
5 failures	Account lockout	15 minutes
10 failures (cumulative)	Extended lockout + admin notification	1 hour
20 failures (24-hour)	Account disabled + security notification	Manual unlock required

REQ-L001-BR-006: Password requirements (for password reset flow):

Requirement	Specification
Minimum Length	12 characters
Complexity	Uppercase, lowercase, digit, special character
Prohibited	Common passwords, username substring, sequential characters
History	Cannot reuse last 6-12 passwords (role-dependent)

REQ-L001-BR-007: Session parameters:

Parameter	Value	Condition
Session Duration	8 hours	Standard session
Session Duration	30 days	"Remember me" checked
Idle Timeout	30 minutes	All sessions
Absolute Timeout	24 hours	All sessions
Concurrent Sessions	5 maximum	Per user

5.3 MFA Business Rules

REQ-L001-BR-008: MFA enforcement:

- Roles with mandatory MFA SHALL be redirected to MFA enrollment if not configured
- TOTP codes valid for +/- 1 period (90-second window)
- "Trust this device" stores device fingerprint for 30 days
- 3 failed MFA attempts triggers account lockout

REQ-L001-BR-009: MFA recovery:

- 10 single-use backup codes provided at enrollment
- Backup codes are 16 characters each
- Using backup code invalidates that code permanently

6. API Integration Points

6.1 Authentication Endpoints

REQ-L001-API-001: The following API endpoints SHALL be implemented:

Endpoint	Method	Description	Request	Response
/auth/login	POST	Primary authentication	{email, password, remember_me}	{session_token} Or {mfa_required: true}
/auth/logout	POST	Terminate session	{}	{success: true}
/auth/mfa/verify	POST	Verify MFA code	{code, trust_device}	{session_token}
/auth/mfa/backup	POST	Use backup code	{backup_code}	{session_token}
/auth/password/reset-request	POST	Request password reset	{email}	{success: true}
/auth/password/reset	POST	Set new password	{token, password}	{success: true}
/auth/sso/init	POST	Initiate SSO flow	{domain}	{redirect_url}
/auth/sso/callback	POST	Handle SSO response	{assertion}	{session_token}
/auth/session	GET	Get current session	-	{session_data}
/auth/refresh	POST	Refresh session token	{}	{new_expiry}

6.2 Request/Response Specifications

REQ-L001-API-002: Login request format:

```
{
  "email": "user@company.com",
  "password": "SecurePassword123!",
  "remember_me": true,
```

```
"device_fingerprint": "abc123..."
}
```

REQ-L001-API-003: Successful login response (no MFA):

```
{
  "success": true,
  "session": {
    "token": "eyJhbGc...",
    "expires_at": "2026-01-02T08:00:00Z",
    "user": {
      "id": "uuid",
      "email": "user@company.com",
      "primary_role": "BRAND_ADMIN",
      "roles": ["BRAND_ADMIN", "CAMPAIGN_MANAGER"]
    },
    "redirect_url": "/admin/dashboard"
  },
}
```

REQ-L001-API-004: MFA required response:

```
{
  "success": true,
  "mfa_required": true,
  "mfa_token": "temp_token_for_mfa_verification",
  "methods": ["totp", "backup_code"]
}
```

6.3 Error Response Format

REQ-L001-API-005: Error responses SHALL follow structured format:

```
{
  "success": false,
  "error": {
    "code": "AUTH_INVALID_CREDENTIALS",
    "message": "Invalid email or password",
    "details": null
  }
}
```

7. State Transitions

7.1 Login State Machine

REQ-L001-ST-001: The login flow SHALL follow this state machine:

```

[Initial]
|
v
[Form Displayed] --submit--> [Validating Credentials]
|                               |
| (cancel)                     | (invalid)
v                               v
[Cancelled]                     [Error Displayed] --retry--> [Form Displayed]
|                               |
|                               (lockout)
|                               v
|                               [Account Locked]

[Validating Credentials]
|
| (valid, no MFA)
v
[Creating Session] --> [Authenticated] --> [Redirecting]

[Validating Credentials]
|
| (valid, MFA required)
v
[MFA Modal] --submit--> [Validating MFA]
|                               |
| (cancel)                     | (invalid)
v                               v
[Form Displayed]               [MFA Error] --retry--> [MFA Modal]
|                               |
|                               (lockout)
|                               v
|                               [Account Locked]

[Validating MFA]
|
| (valid)
v
[Creating Session] --> [Authenticated] --> [Redirecting]

```

7.2 Session Lifecycle States

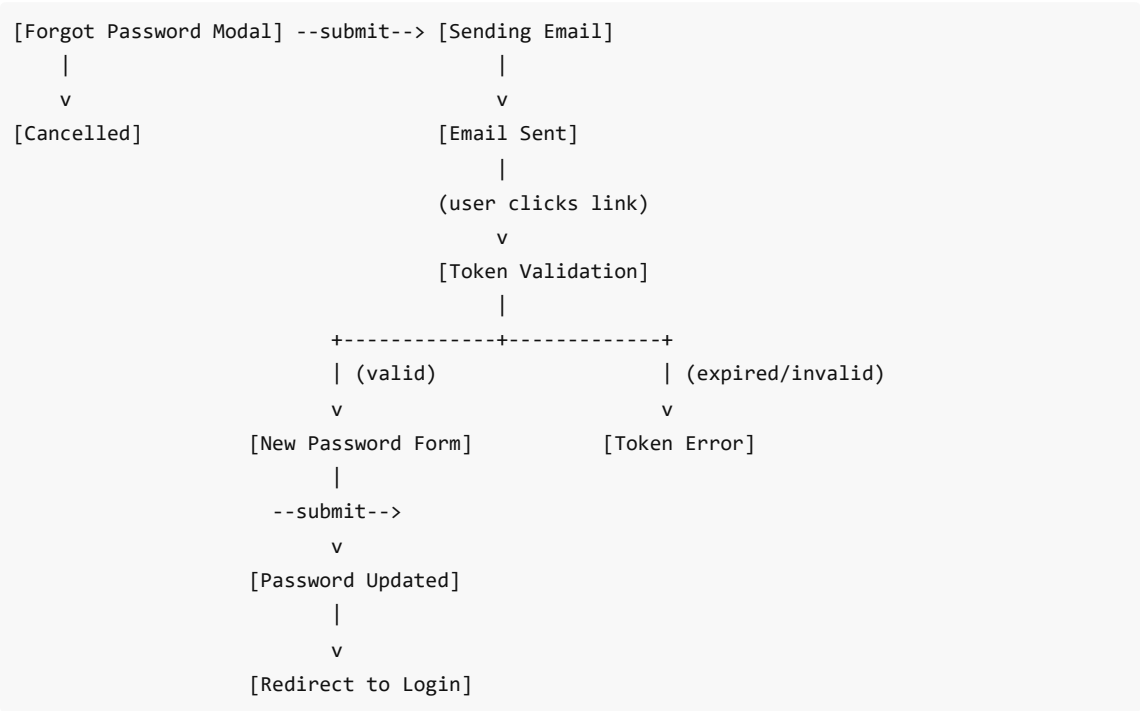
REQ-L001-ST-002: Session state transitions:

State	Triggers	Next States
Created	Successful authentication	Active
Active	User activity	Active (refresh idle timer)
Active	In refresh window + refresh request	Refreshed
Refreshed	New expiry set	Active

Active	30 min no activity	IdleExpired
Active	24 hours since creation	AbsoluteExpired
Active	User logout	Terminated
Active	Password/MFA change	Terminated
Active	Admin action	Terminated
Active	Account suspended	Terminated

7.3 Password Reset State Machine

REQ-L001-ST-003: Password reset flow states:



8. Error Handling

8.1 Error Codes and Messages

REQ-L001-ERR-001: Authentication error handling:

Error Code	HTTP Status	User Message	Internal Log
AUTH_INVALID_CREDENTIALS	401	"Invalid email or password"	Log email, IP, timestamp
AUTH_ACCOUNT_LOCKED	403	"Account locked. Please try again in {X} minutes."	Log lockout duration

AUTH_ACCOUNT_DISABLED	403	"Account has been deactivated. Contact your administrator."	Log disabled reason
AUTH_MFA_REQUIRED	200	(Trigger MFA modal)	Log MFA method
AUTH_MFA_INVALID	401	"Invalid verification code"	Log failure count
AUTH_MFA_EXPIRED	401	"Verification code expired. Please try again."	Log code age
AUTH_SSO_FAILED	400	"SSO authentication failed. Please try again."	Log SSO error detail
AUTH_SSO_DOMAIN_NOT_FOUND	400	"SSO is not configured for this domain."	Log domain
AUTH_RATE_LIMITED	429	"Too many attempts. Try again in {X} minutes."	Log attempt count
AUTH_SESSION_EXPIRED	401	"Your session has expired. Please sign in again."	Log session ID
AUTH_PASSWORD_RESET_INVALID	400	"This password reset link is invalid or expired."	Log token status

8.2 Client-Side Validation Errors

REQ-L001-ERR-002: Form validation error display:

Field	Validation	Error Position
Email	Invalid format	Below email field
Email	Empty	Below email field
Password	Empty	Below password field
MFA Code	Not 6 digits	Below code input
SSO Domain	Invalid format	Below domain field

8.3 Network Error Handling

REQ-L001-ERR-003: Network failure handling:

Scenario	User Message	Retry Behavior
Request timeout	"Request timed out. Please try again."	Automatic retry (1x)
Network offline	"No internet connection. Check your network."	Retry on reconnect
Server error (5xx)	"Something went wrong. Please try again later."	Retry after 5 seconds
Service unavailable	"Service temporarily unavailable. Please try again."	Exponential backoff

9. Accessibility Requirements

9.1 WCAG 2.1 AA Compliance

REQ-L001-A11Y-001: Screen SHALL meet WCAG 2.1 Level AA standards:

Criterion	Requirement	Implementation
1.1.1 Non-text Content	All images have alt text	Logo: alt="NewPOPSys Logo"
1.3.1 Info and Relationships	Form fields have associated labels	<label for="email">
1.4.3 Contrast (Minimum)	4.5:1 for normal text, 3:1 for large text	CSS variables with contrast checking
1.4.11 Non-text Contrast	3:1 for UI components	Button borders, input borders
2.1.1 Keyboard	All functions keyboard accessible	Tab order, Enter to submit
2.4.3 Focus Order	Logical focus sequence	Top-to-bottom, left-to-right
2.4.7 Focus Visible	Clear focus indicator	2px outline with offset
3.3.1 Error Identification	Errors identified and described	Error messages with field association
3.3.2 Labels or Instructions	Form fields labeled	Visible labels, placeholder text
4.1.2 Name, Role, Value	ARIA roles and states	aria-invalid, aria-describedby

9.2 Keyboard Navigation

REQ-L001-A11Y-002: Keyboard interaction requirements:

Key	Action
Tab	Move focus forward through interactive elements
Shift+Tab	Move focus backward
Enter	Submit form / Activate button
Escape	Close modal dialogs
Space	Toggle checkbox / Activate button

9.3 Screen Reader Support

REQ-L001-A11Y-003: Screen reader announcements:

Event	Announcement
Page load	"NewPOPSys Login. Sign in to your account."
Validation error	"Error: {field name} - {error message}"
Form submission	"Signing in..."

Authentication error	"Error: {error message}"
MFA modal open	"Two-factor authentication required. Enter 6-digit code."
Successful login	"Sign in successful. Redirecting..."

9.4 ARIA Attributes

REQ-L001-A11Y-004: Required ARIA implementations:

```
<!-- Email field with error -->
<label for="email">Email Address</label>
<input
  id="email"
  type="email"
  aria-required="true"
  aria-invalid="true"
  aria-describedby="email-error"
/>
<span id="email-error" role="alert">Please enter a valid email address</span>

<!-- Password field -->
<label for="password">Password</label>
<input
  id="password"
  type="password"
  aria-required="true"
/>
<button
  type="button"
  aria-label="Show password"
  aria-pressed="false"
>
  [eye icon]
</button>

<!-- Login button with loading state -->
<button
  type="submit"
  aria-busy="true"
  aria-label="Signing in, please wait"
>
  <span class="spinner" aria-hidden="true"></span>
  Signing In...
</button>
```

10. Acceptance Criteria

10.1 Functional Acceptance Criteria

ID	Criterion	Priority
AC-L001-01	Single login page serves all web portals (Brand Admin, Store Portal, PSP Operations, Regional Dashboard)	Must Have
AC-L001-02	Valid email format validation with inline error display	Must Have
AC-L001-03	Password field shows/hides with visibility toggle	Must Have
AC-L001-04	Successful login with valid credentials redirects to role-appropriate dashboard	Must Have
AC-L001-05	Invalid credentials display generic error message (no credential enumeration)	Must Have
AC-L001-06	MFA modal appears when user has MFA enabled	Must Have
AC-L001-07	Valid MFA code completes authentication	Must Have
AC-L001-08	Invalid MFA code shows error and allows retry	Must Have
AC-L001-09	"Trust this device" option bypasses MFA for 30 days	Should Have
AC-L001-10	Backup code authentication works when MFA device unavailable	Must Have
AC-L001-11	SSO option redirects to identity provider for configured domains	Should Have
AC-L001-12	Forgot password flow sends reset email within 60 seconds	Must Have
AC-L001-13	Password reset link expires after 1 hour	Must Have
AC-L001-14	"Remember me" extends session to 30 days	Should Have
AC-L001-15	5 failed login attempts trigger 15-minute lockout	Must Have
AC-L001-16	Lockout countdown displays remaining time	Should Have
AC-L001-17	Multi-role users see role selector modal	Should Have
AC-L001-18	Session persists across page refreshes	Must Have

AC-L001-19	Logout clears session and redirects to login	Must Have
AC-L001-20	Return URL parameter redirects after authentication	Should Have

10.2 Security Acceptance Criteria

ID	Criterion	Priority
AC-L001-SEC-01	Passwords transmitted over HTTPS only (TLS 1.2+)	Must Have
AC-L001-SEC-02	CSRF token included in login form	Must Have
AC-L001-SEC-03	Session cookie set with HttpOnly, Secure, SameSite=Strict	Must Have
AC-L001-SEC-04	Password hashed with bcrypt (cost factor 12)	Must Have
AC-L001-SEC-05	Rate limiting blocks excessive requests (429 response)	Must Have
AC-L001-SEC-06	Failed login attempts logged with IP address	Must Have
AC-L001-SEC-07	Successful login events logged to audit trail	Must Have
AC-L001-SEC-08	Session ID regenerated on authentication	Must Have
AC-L001-SEC-09	No credential enumeration via error messages	Must Have
AC-L001-SEC-10	Password reset invalidates all active sessions	Must Have

10.3 Performance Acceptance Criteria

ID	Criterion	Target
AC-L001-PERF-01	Page initial load time	< 2 seconds (3G network)
AC-L001-PERF-02	Login API response time	< 500ms (P95)
AC-L001-PERF-03	MFA verification response time	< 300ms (P95)
AC-L001-PERF-04	Session validation response time	< 100ms (P95)
AC-L001-PERF-05	Time to Interactive (TTI)	< 3 seconds

10.4 Accessibility Acceptance Criteria

ID	Criterion	Priority
AC-L001-A11Y-01	Page passes WAVE accessibility checker with no errors	Must Have
AC-L001-A11Y-02	All form fields have visible labels	Must Have
AC-L001-A11Y-03	Color contrast meets WCAG 2.1 AA (4.5:1 normal, 3:1 large)	Must Have
AC-L001-A11Y-04	Complete keyboard navigation without mouse	Must Have

AC-L001-A11Y-05	Screen reader announces form errors	Must Have
AC-L001-A11Y-06	Focus trap works correctly in modals	Must Have
AC-L001-A11Y-07	Escape key closes modal dialogs	Must Have

11. Traceability Matrix

Requirement ID	Description	Source Document	Test Case
REQ-L001-FR-001	Role-based routing priority	SUPP-003, L01 Screen Spec	TC-L001-001
REQ-L001-FR-002	Multi-role handling	SUPP-003, L01 Screen Spec	TC-L001-002
REQ-L001-UI-001	Responsive layout	L01 Screen Spec	TC-L001-010
REQ-L001-UI-002	MFA modal display	L01 Screen Spec	TC-L001-011
REQ-L001-UI-003	Password reset modal	L01 Screen Spec	TC-L001-012
REQ-L001-UI-004	SSO domain entry	L01 Screen Spec	TC-L001-013
REQ-L001-UI-005	Role selector modal	L01 Screen Spec	TC-L001-014
REQ-L001-DR-001	User entity fields	3.1 Database Model	TC-L001-020
REQ-L001-DR-002	Membership query	3.1 Database Model	TC-L001-021
REQ-L001-DR-003	Session data structure	4.3 Authentication Flows	TC-L001-022
REQ-L001-BR-001	Email validation	L01 Screen Spec	TC-L001-030
REQ-L001-BR-002	Password validation	L01 Screen Spec	TC-L001-031
REQ-L001-BR-003	MFA code validation	L01 Screen Spec	TC-L001-032
REQ-L001-BR-004	SSO domain validation	L01 Screen Spec	TC-L001-033
REQ-L001-BR-005	Account lockout policy	4.3 Authentication Flows	TC-L001-040
REQ-L001-BR-006	Password requirements	4.3 Authentication Flows	TC-L001-041
REQ-L001-BR-007	Session parameters	4.3 Authentication Flows	TC-L001-042
REQ-L001-BR-008	MFA enforcement	4.2 Permission Matrix	TC-L001-043
REQ-L001-BR-009	MFA recovery	4.3 Authentication Flows	TC-L001-044
REQ-L001-API-001	Authentication endpoints	L01 Screen Spec	TC-L001-050
REQ-L001-API-002	Login request format	3.4 Integration Architecture	TC-L001-051
REQ-L001-API-003	Login response format	3.4 Integration Architecture	TC-L001-052
REQ-L001-API-004	MFA response format	3.4 Integration Architecture	TC-L001-053
REQ-L001-API-005	Error response format	3.4 Integration Architecture	TC-L001-054

REQ-L001-ST-001	Login state machine	L01 Screen Spec	TC-L001-060
REQ-L001-ST-002	Session lifecycle	4.3 Authentication Flows	TC-L001-061
REQ-L001-ST-003	Password reset states	4.3 Authentication Flows	TC-L001-062
REQ-L001-ERR-001	Error codes	L01 Screen Spec	TC-L001-070
REQ-L001-ERR-002	Validation errors	L01 Screen Spec	TC-L001-071
REQ-L001-ERR-003	Network errors	3.2 Application Architecture	TC-L001-072
REQ-L001-A11Y-001	WCAG 2.1 AA	Accessibility Standards	TC-L001-080
REQ-L001-A11Y-002	Keyboard navigation	Accessibility Standards	TC-L001-081
REQ-L001-A11Y-003	Screen reader support	Accessibility Standards	TC-L001-082
REQ-L001-A11Y-004	ARIA attributes	Accessibility Standards	TC-L001-083

12. Cross-References

Reference	Section	Description
3.1 Database Model	Schema	User, Membership, Session tables
3.2 Application Architecture	Frontend	Next.js App Router implementation
3.3 Technology Stack	Auth	bcrypt, TOTP, WebAuthn specifications
3.4 Integration Architecture	API	Authentication endpoint patterns
4.1 Persona Matrix	Users	User classes accessing this screen
4.2 Permission Matrix	RBAC	Role definitions and MFA requirements
4.3 Authentication Flows	Security	Complete authentication specifications
L01 Screen Spec	Source	Original wireframes and specifications
SUPP-003	Foundation	RBAC and Permissions Matrix

13. Revision History

Version	Date	Author	Description
1.0	2026-01-01	System	Initial SRS specification

Document Status: Draft IEEE 830 Compliance: Section 5 - Screen Specifications Last Updated: 2026-01-01